Input & Initial Processing

PyTorch nn.Module

torch.compile
(Inductor Backend)

Inductor-Aware
Synthesis

Preserved:
Extern Library
(cuBLAS/cuDNN)

Regex
Analysis

LLM Codegen → Validate
[retry if fail]

Targeted
Kernel Pool

Native Triton Kernels

Helion DSL Kernels

Baseline NCU Profiling

T iterations

Iterative Multi-Agent Optimization Loop

deterministic

Strategy
Analyst

Strategy Memory (M)

Updates

Planner

Curated updates
feedback

Executor

(K) retry attempts

correctness failures
compile errors

In-conversation feedback

Curated
updates

$s_+ = 1.05$
$s_- = 1.20$

Profiler

Updates

Summarizer

Selection: DiverseSelect*(C, B)* Algorithm 1

Phase 1: Exploration → Phase 2: Exploitation

global latency comparison *(do_bench)*

Optimized or Fallback compile baseline k_0

Reassembling & Verification

Full compiled model context restitching

Optimized targeted kernels with preserved library calls and full model artifact

per-candidate

Gate 1: Static validation

check: Syntax Error, Runtime Crashes

AND

per-candidate

Gate 2: Multi-seed correctness

allclose(rtol=atol=1e-3)
3 seeds, per-candidate (every iteraton)

AND

end-to-end

Gate 3: Model-level correctness

$$\rho = \frac{\|o_{f32} - r_{f64}\|_\infty}{\|r_{f32} - r_{f64}\|_\infty}$$

Weight extracted E2E verification,
allclose(rtol=atol=1e-4), float64 fallback

AND

end-to-end

Gate 4: Performance gate

$$V_{\text{perf}}(k, k_0) = \mathbb{1}[\tau_M(k_0 \to k) \le \gamma \cdot \tau_M(k_0)]$$

$\gamma = 1.03$